Oriented guards and smaller Diophantine consistency equations

OpenAI-assisted research for Vladimir Reshetnikov and ProveIt

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Abstract

This iteration combines six newly delivered constructions and also investigates architectures that do not assume K215 or Jones as a required starting point. The principal new structural result is a quartic in42 unrestricted integer unknowns, improving45: a cofinal factor-four square scale forces a large modulus before Pell rank recovery, allowing two positivity guards to be replaced by one. Exact monic projections give36-integer sextic and32-integer octic companions. A smaller degree-separated source gives a nine-positive polynomial of degree 1 731 700 in462 operations and a lower-degree 1 494 006/487 tradeoff. Congruence norms and a two-bit carry trim give ten integers, degree 3 327 556, in519 operations. An adaptive divisor modulus gives fourteen positive unknowns of degree30 in87 operations. All are explicit fixed polynomials with complete coefficient bindings. The proofs are ordinary mathematics formalizable in ZFC, with identified inherited arithmetic/compiler dependencies and finite exact certificates; no consistency decision, global optimum, or new end-to-end kernel proof is claimed.

1 Problem, scope and immutable baseline

We seek an explicit polynomial F with integer coefficients and a stated integer domain D for which ordinary mathematics establishes

$$\mathrm{ZFC} \vdash \big(\neg\exists \mathbf{x} \in D^n \; F(\mathbf{x}) = 0\big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Here $\mathbb{N}$ includes zero; strictly positive and unrestricted integer domains are distinguished everywhere. Intermediate registers are polynomial abbreviations, not hidden quantified unknowns. Fixed numerals are supplied as actual payloads or explicit finite integer recipes, not selected according to whether consistency holds.

The six original incoming archives were retrieved at `7331974b6966`, placed at `beb0192cb8dc`. Full immutable IDs, Git blob hashes, archive SHA256 values and byte counts appear in `INCOMING_PINS.json`. The project is

https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/DiophantineConsistency/con-zfc-diophantine-forms.

All six original root replays passed in fresh copies; the original trees were left unchanged. The input packets are Free Coordinate, Integral Root, Separated Degrees, our preceding Degree Separated packet, Calibrated Carry, and Congruence Norms. We use their actual archived circuits and proofs, not a possibly incomplete placed tree. Independently recomputed live-gate counts and the reviewed formal-status snapshot accompany the report.

Neither K215 nor Jones is a requirement of the problem. We considered direct proof predicates, other constructive DPRM compilers, and fixed-input universality. Those alternatives are discussed separately; none is passed off as a validated smaller endpoint. The successful low-degree quartic and its projections do not use the Jones universal expression or Sun's reduction, although their source calibration still uses the established native compiler.

2 Measured results and honest tradeoffs

Binary additions, subtractions and multiplications are charged, including multiplication by a numeral and gates implementing fixed powers of variables. Shared registers are reused; fixed coefficient leaves are free in this circuit metric. Coefficient storage and bit complexity remain separate measures.

Domain, unknowns	Exact degree	Operations	Comparison
9 positive, shorter	1 731 700	462	Previous 1 910 404/473
9 positive, lower degree	1 494 006	487	Degree/cost tradeoff
10$\mathbb{Z}$	3 327 556	519	Previous 8 469 528/533
13$\mathbb{Z}$, no Sun2026	10 461 744	519	Previous 11 543 632/518
14 positive	30	87	Previous 52/79
42$\mathbb{Z}$	4	not optimized	Incoming 45$\mathbb{Z}$, degree4
36$\mathbb{Z}$	6	not optimized	Exact monic projection
32$\mathbb{Z}$	8	not optimized	Exact monic projection

The twelve- and thirteen-positive incumbents 242/94, 258/89 and 126/91, 132/83 are not displaced. The fifteen-positive degree24/90 and 26/88 forms also remain useful. The new fourteen-positive degree30 form saves one variable against the incoming fifteen-positive degree30/79 point at eight additional operations. Its five-residual system costs78 operations and has maximum degree15.

The nine-positive family additionally includes the exact tradeoffs 1 663 788/475, 1 595 876/478 and 1 527 964/482. Natural-coordinate translations add seven gates, so the two highlighted versions cost469 and494. The thirteen-integer result improves degree at one extra gate, not both metrics. Calibrated Carry still supplies a larger-degree ten-integer point with513 operations. The fifteen-positive 66-operation degree157 form remains the absolute operation-count incumbent among these compared families. There is no single winner under every size objective and no global minimality conclusion.

What is genuinely new. The quartic improvement uses cofinal witness flexibility to establish $|E| > 3R + 1$ before rank recovery; it is not merely a gate optimization. The source improvement combines incoming108-position weights with the previous row shear, fresh target placement and a global sign reversal. All three reversed kernels now have positive leaders, simplifying the unconditional positivity argument. The nine-variable alternatives exploit higher Pell majorants while retaining positivity-first free-coordinate soundness. The ten-integer construction combines the smaller source with incoming two-root congruence separation and removes two provably redundant population bits. The fourteen-positive construction uses the recovered scale divisibility to replace a fixed modulus by $4qY$.

Coefficient coherence. The small-variable source families retain the whole original31 tuple, with native radix $2^{1064921}$; no one port is borrowed from a smaller adapter. The positive14 family instead uses its entire inherited compact 2^{590069} tuple. The quartic uses the complete package34 mixed-parity-unrounded tuple at radix $2^{1064921}$, which must not be identified with the different original31 ports merely because the radix exponents match. Every binding supplies the already shifted high mask once. No new coefficient-storage record is claimed.

3 A quartic in forty-two unrestricted integers

The new equation has **42 unrestricted integer unknowns**, exact degree four, and 27 quadratic residuals. It improves the incoming 45-integer, 28-residual quartic without relying on the Jones universal expression or Sun's ten-integer reduction. Its semantic source remains the inherited native compiler and ZFC-refutation calibration. The new argument is a cofinal scale restriction that orients a shared sign guard before any power-of-two recovery.

3.1 The complete quadratic system

Use the 32 arithmetic variables

$$J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B, \Delta, u,$$

one additional variable q_h, and three triples $\mathbf{t}_A, \mathbf{t}_C, \mathbf{t}_K$. All42 range over $\mathbb{Z}$. The letters B, C_2, F_2 are independent lifted registers, not radicals. The four fixed numerals (β, K_0, M_C, M_F) are the entire retained package34 tuple, $\beta = 2^{1064921} - 1$; M_F is already shifted. Define only polynomial abbreviations

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad \mathrm{Pos}(x, y, t) = 1 + x^2 + y^2 + t(t+1),$$
$$\widehat{R} = F(q + L - N) + J(M_C + qM_F) + q(N - L - q) + L\alpha.$$

On the first two equations, $L = q^2 - 1$, $N = q^3$ and $\widehat{R} = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J$. The 27 residuals are the following explicitly quadratic expressions:

$$e_0 = L - q^2 + 1, \qquad e_1 = N - q(L+1),$$
$$e_2 = X - qw, \qquad e_3 = a - E - (2v+1)N,$$
$$e_4 = E(X+1) - Xa, \qquad e_5 = c - k(a - E) - \eta,$$
$$e_6 = P - E(c - \eta) - k, \qquad e_7 = \tau^2 - P^2 + Pk - 1,$$
$$e_8 = D - X - ac - (4a+3)\gamma, \qquad e_9 = \Delta - a^2 - 4a - 3,$$
$$e_{10} = C_2 - c^2, \qquad e_{11} = D^2 - \Delta C_2 - 1,$$
$$e_{12} = k - 1 - Eh - \widehat{R},$$
$$e_{13} = (K_0 + w)(q - F - \alpha) - F - z(q-1),$$
$$e_{14} = F_2 - f^2, \qquad e_{15} = V + c - fo,$$
$$e_{16} = V - cj + k - 1 - Eh, \qquad e_{17} = V^2 + g - y^2,$$
$$e_{18} = I - (2r+1)C_2, \qquad e_{19} = B - I^2,$$
$$e_{20} = B - \Delta F_2 + \Delta, \qquad e_{21} = y^2 - Bg - 1,$$
$$e_{22} = w - 4u^2, \qquad e_{23} = q - 2q_h^2,$$
$$e_{24} = F\alpha - \mathrm{Pos}(\mathbf{t}_A),$$
$$e_{25} = F(q - F - \alpha) - \mathrm{Pos}(\mathbf{t}_C),$$
$$e_{26} = \eta\zeta - \mathrm{Pos}(\mathbf{t}_K).$$

The single recommended equation is

$$\boxed{Q_{42} = \sum_{j=0}^{26} e_j^2 = 0.}$$

The supplied exact expansion has391 symbolic terms. The coefficient of τ^4 is1, so specializing the fixed numerals cannot reduce its degree. All variables occur; no positivity side condition is left implicit.

3.2 What changes from the forty-five-variable source

The predecessor guards η and ζ separately using six ternary positivity variables. We replace those equations by $e_{26} = 0$, retaining only three such variables. This initially establishes only that η and ζ have the same nonzero sign. At the same time, we replace its $w = u^2$ by $w = 4u^2$, adding no variable or residual. Hence the counts are $45 - 6 + 3 = 42$ and $28 - 2 + 1 = 27$.

Every positive integer is represented by Pos, by the three-square theorem applied to $4n - 3$. The restriction $w = 4u^2$ is not asserted for every predecessor solution: completeness will use the canonical cofinal family. Omitting this strengthened scale leaves a real negative-common-sign branch and does not give our proof.

3.3 Recover the common sign before invoking the source theorem

At a zero of Q_{42} every e_j vanishes. The dyadic guard gives $J > 0$ and $q \geq 16$. The two simplex product guards, together with $F + \alpha + (q - F - \alpha) = q > 0$, give $F, \alpha, q - F - \alpha > 0$. The unchanged mask bounds give

$$q(q^2 - 1) < R := \widehat{R} < q^4 - 2.$$

They require no sign of z, h, v or k. The arithmetic lift implies

$$s = 2v + 1, \quad Y = sq^3 = a - E, \quad E = XY, \quad c = kY + \eta, \quad a = (X + 1)Y.$$

Thus $Y \neq 0$, $Z = |Y| \geq q^3$. If $u = 0$, then $X = 0$ and reducing the main Pell norm modulo $H = 4a + 3$ gives $0 \equiv 1 \pmod{H}$, impossible since $|H| > 1$. Therefore $u \neq 0$, and the strengthened scale proves immediately

$$X \geq 4q, \qquad |E| \geq 4q^4 > 3R + 1. \tag{Q1}$$

This precedes the rank argument, so it is not inferred from an eventual identity $X = 2^R$.

Write $\sigma = \operatorname{sign}(k)$, $k_0 = |k|$, $c_0 = |c|$, $A_0 = |a + 2|$ and $P_0 = 2XY^2 + 1$. Since $\eta\zeta > 0$, $0 < |\eta| < k_0$; hence

$$k_0(Z - 1) < c_0 < k_0(Z + 1), \quad c_0 > k_0, \quad P_0 > A_0 > |E| > R.$$

The consecutive-product and main Pell classifications give $n, p \geq 1$ with

$$k_0 = 2\psi_{P_0}(n), \quad c_0 = \psi_{A_0}(p), \quad p > n.$$

The index equation and $P_0 \equiv 1 \pmod{|E|}$ yield

$$2n \equiv \sigma(R + 1) \pmod{|E|}.$$

For $\sigma = 1$, $n \geq (R + 1)/2$. For $\sigma = -1$, $n \geq (|E| - R - 1)/2 > R$ by(Q1). Both give $n \geq 24$, so $p \geq 25$ and standard Pell growth proves

$$c_0 > A_0\Delta^2, \qquad c_0 > 2p, \qquad c_0 > 2R. \tag{Q2}$$

The ordinary strong-rank lemma is therefore applicable in both sign branches, not just after assuming $k > 0$.

Put $f_0 = |f|$ and $i = 2r + 1 \neq 0$. The retained strong equation gives a rank m with $f_0 = \chi_{A_0}(m)$, $p \mid m$, $c_0 \mid m$ and $m > 2p$. The auxiliary norm and the two exposed quotient equations give

$$V \equiv -c \pmod{f_0}, \qquad V \equiv -R \pmod{c_0}.$$

The odd-Chebyshev quotient identities and signed step-down give $R \equiv \pm p \pmod{c_0}$; squaring the first congruence removes the sign of c. The argument is unchanged for either sign of Y. The strict bounds(Q2) force $p = R$. If $\sigma = -1$, this contradicts $p > n > R$. Consequently $k > 0$ and $\eta, \zeta > 0$.

Only now invoke the predecessor's complete soundness theorem: keep its arithmetic coordinates, put $u_{\text{old}} = 2u$, and represent the now-positive η and ζ by their old ternary guards. Every old residual vanishes literally. The predecessor thus supplies a finite ZFC refutation. Its additional negative-scale and signed-root analyses need not be repeated.

3.4 Cofinal completeness and exact scope

Conversely, its canonical refutation witnesses have $q = 2^t$ with t odd, $R \equiv 3 \pmod 4$, and $w = 2^{R-t}$. Since $R > 3q$, $R - t$ is even and at least2. Choose $u = 2^{(R-t-2)/2}$; then $w = 4u^2$. Represent $\eta\zeta > 0$ by the new ternary guard and retain the other canonical coordinates. All27 residuals vanish.

This proves both directions without deciding consistency. The retained Q45, ordinary-rank, signed step-down and native-calibration sources are identified and hashed in the archive. An independent symbolic expansion checks the exact residual changes and degree. Those finite checks support the literal expression; they do not replace the unbounded inherited arithmetic/compiler theorems or constitute a new kernel certificate.

3.5 Exact monic projections: thirty-six and thirty-two integers

A residual $x - P(\mathbf{y}) = 0$ with coefficient1 permits deleting x and substituting its integer polynomial value in every other residual. This is an exact bijection of integer solution sets, with no divisibility side condition. Acyclic chains of these substitutions preserve the same consistency equivalence.

For the sextic, eliminate the six lifted coordinates

$$w = 4u^2,\ C_2 = c^2,\ F_2 = f^2,\ B = I^2,\ L = q^2 - 1,\ E = a - (2v+1)N.$$

The remaining21 residuals are cubic or lower. Their sum of squares is an explicit polynomial in36 unrestricted integers of exact degree6; the coefficient of $I^4 g^2$ is1. The archive gives its435-term symbolic expansion and the complete reconstruction map.

For the octic, eliminate instead

$$\begin{gathered} w = 4u^2,\ X = qw,\ C_2 = c^2,\ \Delta = a^2 + 4a + 3,\ F_2 = f^2, \\ B = I^2,\ g = y^2 - V^2,\ L = q^2 - 1,\ N = q(L+1), \\ P = E(c - \eta) + \eta + \zeta. \end{gathered}$$

The remaining17 residuals have degree at most4, so their sum of squares is an octic in32 unrestricted integers. Its coefficient of $I^4 V^4$ is1; its exact expansion has395 symbolic terms. The substitutions $X = qw$ and $N = q(L+1)$ are fully expanded after those for w and L, respectively. Neither eliminated register remains an existential unknown.

These are degree–variable tradeoffs, not42-variable quartics relabeled with smaller counts. The supplied builders independently check the monic identities, acyclic reconstruction, absence of eliminated symbols, degrees and unit leading coefficients. Their semantic justification is the elementary integer projection lemma above, composed with the proved Q42 reduction.

4 A positive three-kernel source of height 1954

The new source combines four ingredients: the 108-position weights of the incoming *Separated Degrees* construction, our earlier two row shears, a new target placement, and a global reversal of sign. The last operation makes all three kernel leaders positive. The source has 89 records and convolution height 1954, versus 89 records and height 2161 in our previous source. These are source-code parameters, not additional Diophantine unknowns.

Fixed parent and scope. Use the original 31-positive-coordinate parent, whose SHA-256 is recorded in `SOURCE_PINS.json`, and its complete original31 coefficient tuple. Its ports β, K_0, M_C, M_F have bit lengths $1064921, 1070004, 1064921, 1064922$, with $\beta = 2^{1064921} - 1$ and the already shifted M_F. No port from a different compiler tuple is substituted. The parent-to-refutation compiler theorem is inherited; the present finite certificate proves algebraic extraction identities, not that unbounded theorem.

Row basis and exact kernels. If r_j are the 22 parent residuals, put

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3, \qquad \rho_j = r_j\ (j \neq 14, 15), \qquad \sigma_j = -\rho_j.$$

These invertible changes preserve the common zero set. In parent coordinate order the weights are

$$\begin{aligned} (w_i) = (&104, 76, 79, 60, 108, 8, 99, 55, 36, 54, 98, 11, 7, 10, 44, 32, \\ &106, 22, 17, 1, 3, 4, 37, 89, 97, 107, 39, 40, 86, 45, 92). \end{aligned}$$

The row order and corresponding targets are

$$\begin{aligned}(j) &= (8,15,20,9,4,19,7,12,13,17,16,10,5,21,1,3,11,6,14,18,22,2),\\ (q_j) &= (34,229,274,319,524,533,573,767,820,1007,1053,1064,\\ &\quad 1089,1291,1320,1366,1381,1410,1617,1707,1798,1939).\end{aligned}$$

For a degree-d monomial ap^α of σ_j, place $2a\alpha!/d!$ at exponent $q_j - \sum_i \alpha_i w_i$ of $D_d(T)$. Only $d = 0, 1, 2$ occur. With $U_p(T) = \sum_i p_i T^{w_i}$, the exact certificate gives

$$[T^{q_j}]\big(D_2(T)U_p(T)^2 + D_1(T)U_p(T) + D_0(T)\big) = 2\sigma_j(p).$$

All kernels are explicit in `split_kernels.json`; global negation is already applied there. Their degrees are $1939, 1842, 1738$ and leading coefficients $2, 2, \beta$, respectively. Consequently

$$W = 108, \quad Q = 1939, \quad H = \max_d(\deg D_d + dW) = 1954.$$

The checker independently parses the parent residuals, verifies the inverse shears, and checks 22,544 isolation interactions yielding all 22 identities.

A simpler unconditional sign proof. Every kernel coefficient has absolute value below $C = 2^{2134927}$. Fix $\kappa = 2134940$ and set

$$B = 2^\kappa b^2 = 8192Cb^2, \quad \ell = \sum_i B^{w_i}, \quad U = \ell + g, \quad G = D_2(B)U^2 + D_1(B)U + D_0(B).$$

For a kernel with degree h and positive leader a, the absolute lower-tail contribution is less than $CB^h/(B-1)$. Since $B > 2C$, each kernel is positive. Thus $G > 0$ whenever $b, g \geq 1$, without a binary-base assumption or a final-equation hypothesis. No cancellation between the three kernels is needed.

Define

$$A_* = \sum_j B^{q_j}, \qquad N_0 = B^{109}, \qquad Z = B^{1955},$$
$$M = N_0 - 1 - (b-1)\ell, \qquad J_* = 2G + A_*,$$
$$S = g + N_0 J_*, \qquad T = M + N_0(B-2)A_*, \qquad G_{\mathrm{cap}} = 32bB^{108}.$$

Then $S > 0$ and $T \geq 0$ unconditionally. On the ratio-derived range $g < 4bB^{108}$, one has $U < 6bB^{108}$ and

$$G < \frac{C}{B-1}\big(36b^2B^{1955} + 6bB^{1951} + B^{1940}\big), \qquad \frac{2G}{Z} < \frac{86Cb^2}{B-1} < \frac{1}{64}.$$

Also $A_* < Z/64$, $(B-2)A_* < Z/64$, and $g < N_0$. Hence

$$S + T < N_0 Z = B^{2064}.$$

This is proved on the full necessary range, not merely on the smaller range occupied by canonical witnesses.

Signed-digit decoding without spurious zeros. For $b = 2^k$ and $B = 2^v$, the low mask M has digit $B - b$ at each of the 31 active positions, and digit $B - 1$ elsewhere below position109. Thus disjointness of g and M is equivalent to $g = \sum_i z_i B^{w_i}$ with $0 \leq z_i < b$. The word U then encodes $p_i = z_i + 1 > 0$. Every convolution coefficient h_t satisfies

$$|h_t| < C\big((\textstyle\sum_i p_i)^2 + \sum_i p_i + 1\big) \leq 993Cb^2 < B/8.$$

The formal coefficients of J_* are $2h_t + \mathbf{1}_{\{t \text{ target}\}}$. Base-$B$ normalization therefore uses only incoming borrows $0, -1$. At a target, $h_t = 0$ produces digit 1 or 0, both disjoint from $B-2$. For $h_t > 0$ the digit lies between 2 and $B/2$; for $h_t < 0$ it lies between $B/2$ and $B-1$. Both nonzero cases intersect $B-2$. Consequently

$$S \,\&\, T = 0 \quad \Longleftrightarrow \quad g \text{ encodes a positive parent zero.}$$

Conversely every parent zero is encoded for all sufficiently large binary b, with $0 < g < bB^{108}$ and $32g < G_{\text{cap}}$. The all-one parent tuple is not a zero, since $r_5(1, \ldots, 1) = 1$; this proves the strict lower bound on g. Every weight is positive, so canonical g is divisible by B and hence by b^2.

Exact recalibration for the free-coordinate finalizer. Writing $v = \kappa + 2k$, the mask population is

$$\operatorname{pop}(T) = 131v - 31k - 22.$$

Set

$$L = bB^{2064}, \quad R = (S+T)L + L - 1 - S, \quad c = 2^{15\kappa/2-11} = 2^{16012039}, \quad N = cB^{1090}.$$

The even choice of κ ensures that c is an integer. Unconditionally $R = (L-1)S + LT + L - 1 \geq 8$. On the bounded branch $S + T < L$; if $u = \log_2 L = k + 2064v$, then

$$\operatorname{pop}(R) = u + \operatorname{pop}(T) - \tau, \qquad \tau = \operatorname{pop}(S) + \operatorname{pop}(T) - \operatorname{pop}(S+T) \geq 0.$$

Here $\tau = 0$ exactly when $S \,\&\, T = 0$, and

$$u + \operatorname{pop}(T) = 2195v - 30k - 22 = 2\log_2 N.$$

Legendre's valuation identity proves $N^2 \mid \binom{2R}{R}$ exactly when the source is satisfied. The degrees entering the arithmetic finalizer are $\deg R = 8255$ and $\deg N = 2180$; intermediate circuit registers are not additional variables.

Alternative integer-route calibration. For $n = B^{2064}$ put

$$X = n^2 + (S+T)n + n - 1 - S, \qquad Y = 2^{2195\kappa-21} b^{4359}.$$

On the bounded binary branch, $\operatorname{pop}(X) = \log_2 Y - \tau$. When $b \equiv 1 \pmod 3$, even κ gives $B \equiv 1 \pmod 3$, $T \equiv 2 \pmod 3$ and $X \equiv 0 \pmod 3$. Also $X + 1 \equiv -g \pmod b$. These are source interface facts; the separate integer finalizer must still prove its numerical and integrality obligations.

Reproduction and contribution boundary. Run `python verify_split.py`, `python test_split.py`, and `python check_carry.py` from the source directory. The isolated replay also passes under Python `-O`; eleven intentional corruptions are rejected. Carry sanity tests accompany, but do not replace, the general argument above. The borrowed weights, inherited compiler, sparse borrow method and population theorem are credited in the provenance files. The new contribution is their compatible combination with the shears, placement, global sign and exact recalibration. No global minimum, consistency proof, or new end-to-end kernel certificate is asserted.

5 Nine-variable equations: positive signed kernels and tunable norms

The new source encoding and the free-carrier argument combine without any additional unknowns. For positive integer unknowns $b, g, h, s, w, C, i, j, n$, the following equations have no positive integral zero if and only if Con(ZFC), provably in ZFC, using the inherited source/compiler and numerical ratio theorems.

Majorant order	Exact degree	Binary arithmetic operations
Baseline	1,731,700	462
1	1,663,788	475
2	1,595,876	478
3	1,527,964	482
4	1,494,006	487

The baseline improves both measures over the previous 1,910,404/473 construction. The other rows are degree/cost tradeoffs, not claims to dominate on every measure. An order-zero variant has degree 1,731,700 and 465 operations and is retained only for checking.

5.1 The source and numerical interface

Use the unchanged original31 four-coefficient tuple, supplied as binary payloads, and the signed split kernels of the preceding construction. Their degrees are $1939, 1842, 1738$, with positive leading coefficients $2, 2, \beta$. For $M = 2^{2134927}$, every coefficient has absolute value less than M, so each kernel is positive when $B > 2M$. Let $\kappa = 2134940$, $a = 2^\kappa$, $B = ab^2$, and define

$$\ell = \sum_i B^{w_i}, \qquad U_s = \ell + g, \qquad G_s = (D_2(B)U_s + D_1(B))U_s + D_0(B),$$
$$Q_s = \sum_j B^{q_j}, \qquad N_0 = B^{109}, \qquad T_s = N_0 - 1 - (b-1)\ell + N_0(B-2)Q_s,$$
$$S_s = g + N_0(2G_s + Q_s), \qquad L = bB^{2064}, \qquad N = 2^{16012039}B^{1090},$$
$$R = L(S_s + T_s + 1) - S_s - 1, \qquad t = bB^{108}.$$

The maximum weight is 108, convolution height is 1954, and the maximum target is 1939. The global sign reversal preserves the source zero set. The finite source certificate verifies all 22,544 isolation interactions. For binary $b = 2^k, B = 2^v$, the calibrated threshold is

$$\operatorname{pop}(R) = 2195v - 30k - 22 - \delta = 2\log_2 N - \delta,$$

where $\delta \geq 0$ is precisely the carry defect. The even value of κ matters: N has no separate factor b.

Set $j_0 = j - 1$ and define the shared ratio registers

$$U = N^2 w, \qquad Y = N^2 s, \qquad M_0 = RY,$$
$$P = 2M_0^2 U, \qquad K = R + 1 + h(P-1), \qquad A = M_0(U+1),$$
$$B_1 = 2R + 1, \qquad W_1 = bw, \qquad J = 4A - 5,$$
$$Q = 3W_1 C - 2(W_1^2 - 1), \qquad D = (A^2-1)C^2 + 1, \qquad e = iJC,$$
$$E = eDC, \qquad F = (A^2-1)E^2 + 1, \qquad G = A + F(CD + 1 - A),$$
$$H = C + F(B_1 + j_0 C), \qquad I = (G^2-1)H^2 + 1,$$
$$a_0 = (P^2-1)K^2 + 1, \qquad a_1 = DFI, \qquad V_0 = PK,$$
$$m = K^2(4t - g) - 16t(C - KY)^2, \qquad \Omega = Jm, \qquad E_0 = J(n + W_1^2) + Q.$$

The original numerical interface becomes exactly: a_0, a_1 are squares, $J \mid Q$, and $m > 0$. The latter recovers $C > K(Y - 1/2) > K$ and $C > B_1 + W_1$; it is not an assumed domain restriction. The exact old-index correspondence is $j_{\text{old}} = 1 + ((F-1)/C)B_1 + Fj_0$.

5.2 Higher majorants without extra congruence restrictions

The baseline takes $\lambda = 8$ and $T = e(8D^2 - 4D - 1)G$. For order $r = 0, 1, 2, 3, 4$, put $s_r = 128, 64, 8, 4, 1$, respectively, and $\lambda = 2048/s_r$. Define

$$L_m = CD + 1 - A, \qquad N_* = 2G + L_m,$$
$$N_1 = 2e^2 D(D-1)L_m + 3C, \qquad N_2 = 2e^2(D-1)L_m,$$
$$N_3 = 2e^2(L_m - C), \qquad N_4 = 2e^2 C.$$

With $(c_1, c_2, c_3, c_4) = (64, 40, 28, 21)$, let

$$T = e\left[\frac{1024D^2 - 512D - 128}{s_r} N_* - \sum_{j=1}^{r} \frac{c_j}{s_r} N_j\right] + \epsilon_r, \qquad V_1 = TH,$$

where $\epsilon_0 = \lambda$, $\epsilon_r = 512i/s_r + \lambda$ for $1 \le r \le 3$, and $\epsilon_4 = 256i(e^2 + 2) + 2048$. All displayed divisions are fixed integer coefficients.

For all these choices, $V_1 > \lambda\sqrt{a_1}$ holds unconditionally. Here is a finite proof rather than an asymptotic series argument. Put

$$t_r = \frac{1024D^6 - 512D^5 - 128D^4 - \sum_{j=1}^{r} c_j D^{4-j}}{s_r}.$$

The exact polynomial $t_r^2 - (1024/s_r)^2 D^{11}(D-1)$ has only positive nonzero coefficients, and $t_r > 0$. The remainders in $N_* = D^j N_j + r_j$ are

$$r_1 = 3 - A, \qquad r_2 = 3CD + 3 - A,$$
$$r_3 = 2e^2(A-1)D^2 + r_2, \qquad r_4 = 2e^2(1 - A - C)D^3 + r_3.$$

For $A \ge 3, C, i \ge 1$, $eA/D < 9i/2$, $eC/D < 3i/2$, and $r_2, r_3 > 0$. Thus orders one through three have adverse remainder less than $288i/s_r$. For order four it is less than $252ie^2 + (765/2)i$. Each correction exceeds this loss plus λ. Finally, with $X = e^2 D^3(D-1)$, strict square-root concavity yields

$$\lambda\sqrt{DFG} < \frac{\lambda}{2} eD^2\sqrt{1 - 1/D}\, N_* + \lambda v, \qquad v = \frac{CD + 1}{2eD\sqrt{1 - 1/D}} < 1.$$

This proves $T > \lambda\sqrt{DFG}$, and hence the required majorant. No parity condition on g, h is needed.

5.3 The equation and its equivalence

The delivered single polynomial is

$$\boxed{[\Theta^2 - \Omega^2(a_0 + \lambda^2 a_1)]^2 - 4\lambda^2\Omega^4 a_0 a_1 = 0, \qquad \Theta = E_0 - \Omega(V_0 + V_1).}$$

All intermediate names are abbreviations, not extra unknowns. Unconditionally $E_0 = Jn + (J-2)W_1^2 + 3W_1 C + 2 > 0$. A zero cannot have $m = 0$. Real factorization gives signed roots $u_0^2 = a_0, u_1^2 = \lambda^2 a_1$ and

$$n + W_1^2 + Q/J = m(V_0 + V_1 + u_0 + u_1).$$

Both strict majorants make the last parenthesis positive, so $m > 0$ follows before any assertion about rational roots. The recovered carrier range implies $a_1 > C^2A^4 > K^2P^2 > a_0$. Consequently $x = u_0 + u_1 = \Theta/\Omega \ne 0$; the identity $2xu_0 = x^2 + a_0 - \lambda^2 a_1$ proves both roots rational, hence integral. Also u_1/λ is integral. The factor identity now forces $J \mid Q$, proving soundness by the inherited ratio theorem.

Conversely use the ratio witnesses, choose positive roots, and set $n = m(V_0 + V_1 + u_0 + u_1) - W_1^2 - Q/J$. It is an integer greater than zero because $W_1^2 + Q/J < 2D < V_1$. This proves completeness and the claimed consistency equivalence. Natural-domain circuits use the bijective shifts, not an unjustified alteration of the CRT index.

5.4 Exact cancellation checks

Writing $\delta_0 = V_0^2 - a_0$, $\delta_1 = V_1^2 - \lambda^2 a_1$, and $B_* = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2(\delta_0 + \delta_1)$, the two-defect identity is

$$F_r = B_*^2 + 4\Omega^2 V_0 V_1 B_* + 4\Omega^4(\delta_0 V_1^2 + a_0\delta_1).$$

Finite remainder identities resolve $D^8\delta_1$, giving a division-free certificate equal to $D^{16}F_r$. The exact degree follows from a multivariate upper bound and a positive nonzero diagonal leading monomial in the actual fixed coefficients. Independent modular leading-coefficient checks use all four materialized payloads. The primary equation never divides by D; its degree is recovered by subtracting $16 \deg D$ from the certificate degree. The package contains the explicit circuits, full proof, payload hashes, symbolic ring checks, source checks, and independent literal replay. These are mathematical proofs and reproducible finite audits, not a new end-to-end kernel certificate or a minimality claim.

6 Degree-separated coding: a ten-integer equation

There is a completely fixed integer polynomial

$$F_{10}(t, g_0, h_0, k_0, l, w, x, y, z, m)$$

of exact total degree 3 327 556, evaluated by a division-free circuit of 519 binary arithmetic operations, for which ordinary mathematics formalizable in ZFC proves

$$\neg\exists(t, g_0, h_0, k_0, l, w, x, y, z, m) \in \mathbb{Z}^{10}\ (F_{10} = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This combines the preceding degree-separated source architecture with the incoming congruence-norm finalizer. Neither ingredient alone is claimed as new. The contribution is their proved compatibility, including the constructive residue restrictions, the smaller source layout, and removal of an irrelevant low mask bit together with the leading high bit of the binomial argument. The inherited compiler calibration and Sun arithmetic bridge remain stated mathematical dependencies. No consistency decision, global minimality claim, or new end-to-end proof-assistant certificate is asserted.

The structural change is to split the extraction convolution by homogeneous degree. A linear label may now collide with a quadratic label: their kernels are separate summands. This relaxes the finite isolation problem rather than merely finding a better placement for the previous problem. Elementary row operations, one shared positivity shift, borrow-tolerant target padding, and exact mask-population calibration complete the construction.

6.1 Fixed source and three homogeneous kernels

Use exactly the unshifted, strictly positive, 31-coordinate parent in `data/parent_quartic31.json`, with its complete original four-port binding $(\beta_{\mathrm{native}}, K_0, M_C, M_F)$. The four included compressed payloads have bit lengths $(1064921, 1070004, 1064921, 1064922)$; the last is `MF_source`, already shifted, and is not shifted again. `INSTANCE.json` is the active binding. No package-52 transport row or coefficient is spliced into this tuple. The pinned parent theorem is

$$\exists p \in \mathbb{Z}_{>0}^{31}\ r_1(p) = \cdots = r_{22}(p) = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con(ZFC)}.$$

Define $\rho_{14} = -(r_{14} + r_2)$, $\rho_{15} = -(r_{15} + r_3)$, and $\rho_j = -r_j$ otherwise. The inverse row operations preserve the common zero set over every coefficient tuple. There are now 89 records: seven constant, 39 linear, and 43 quadratic.

In the parent coordinate order

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta),$$

choose the positive, distinct weights

$$(w_1,\dots,w_{31}) = (104,76,79,60,108,8,99,55,36,54,98,11,7,10,44,32,\\ 106,22,17,1,3,4,37,89,97,107,39,40,86,45,92).$$

The row order and targets are

$$(8,15,20,9,4,19,7,12,13,17,16,10,5,21,1,3,11,6,14,18,22,2),\\ (34,229,274,319,524,533,573,767,820,1007,1053,1064,1089,1291,\\ 1320,1366,1381,1410,1617,1707,1798,1939).$$

Let q_j denote the target assigned to row j, and write $\rho_j = \sum_\alpha a_{j,\alpha} p^\alpha$. For $d = |\alpha|$ put

$$e(\alpha) = \sum_i \alpha_i w_i, \qquad c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & d = 0,1, \\ a_{j,\alpha} \prod_i \alpha_i!, & d = 2. \end{cases}$$

Define the fixed integer polynomials

$$D_d(Z) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} Z^{q_j - e(\alpha)}, \qquad U_p(Z) = \sum_i p_i Z^{w_i}, \qquad G_p(Z) = D_2(Z)U_p(Z)^2 + D_1(Z)U_p(Z) + D_0(Z).$$

All reversed exponents are nonnegative. The exact finite certificate checks, for every used degree-d monomial α and every degree-d monomial γ in 31 variables,

$$q_j - e(\alpha) + e(\gamma) = q_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma.$$

There are respectively $1, 31, 496$ expansion monomials in degrees $0, 1, 2$, and 22,544 source/expansion interactions. There is no requirement to separate a degree-one label from a degree-two label. Multinomial expansion and the displayed coefficient weights therefore give the universal identities

$$[Z^{q_j}] G_p(Z) = 2\rho_j(p) \quad (1 \le j \le 22). \tag{1}$$

The independent checker compares the entire extracted polynomials in all 31 coordinates and four ports, not just evaluations at test points.

Write $M = 108$, $Q = 1939$. The exact kernel degrees and convolution height are

$$\deg D_0 = 1939, \qquad \deg D_1 = 1842, \qquad \deg D_2 = 1738, \qquad H = \max_d (\deg D_d + dM) = 1954.$$

The leading coefficients of (D_0, D_1, D_2) are respectively $(2, 2, \beta_{\text{native}})$, all positive. Every kernel separately has distinct exponents. Replacing each port monomial in a coefficient by its strict bit bound and taking absolute integer coefficients gives a maximum majorant of bit length 2134927. Consequently, with

$$C_0 = 2^{2134927}, \qquad \kappa = 2134940, \qquad \beta_S = 2^\kappa,$$

every kernel coefficient has absolute value less than C_0, and $\beta_S > 4096 C_0$.

6.2 Source polynomials and pre-binary bounds

In ten unrestricted integer unknowns $(t, g_0, h_0, k_0, l, w, x, y, z, m)$ define first

$$f = 2t + 1, \quad b = 6t + 4, \quad g = bg_0, \quad h = bh_0, \quad k = b(3k_0 + 2).$$

The circuit variable names are $(f, g, h, k, l, w, x, y, z, m)$; its first four names denote the free coordinates (t, g_0, h_0, k_0), not these computed registers. Then define

$$b = 1 + 3f, \quad B = \beta_S b^2, \quad \ell = \sum_i B^{w_i}, \quad U_c = \ell + g,$$
$$G_c = D_2(B)U_c^2 + D_1(B)U_c + D_0(B), \quad A_T = \sum_j B^{q_j},$$
$$N_0 = B^{109}, \quad N_1 = B^{1955}, \quad N = N_0 N_1 = B^{2064},$$
$$M_0 = N_0 - 1 - (b-1)\ell, \quad J_c = 2G_c + A_T,$$
$$S_c = g + J_c N_0, \quad T_c = M_0 + (B-2)A_T N_0 - 2,$$
$$X = (S_c + T_c)N + N - 1 - S_c,$$
$$Y = \beta_S^{15}\, 2^{\kappa-23} B^{2179} b = 2^{2195\kappa-23} b^{4359}, \qquad \mathcal{C} = bB^{108} = 2^{108\kappa} b^{217}.$$

All powers have fixed nonnegative integer exponents. Every displayed register is an abbreviation, never an additional quantified unknown.

Assume $f \neq 0$, b is a square, and $0 \leq g < 2\mathcal{C}$. Then $b \equiv 1 \pmod 3$ implies $b \geq 4$, and $B > 4096 C_0 b^2$. Distinct positive weights give $\ell < 2B^{108}$, hence

$$B^{108} \leq U_c < 3bB^{108}.$$

Every lower coefficient has absolute value below C_0. For a kernel of degree d with positive integer leading coefficient c, the sum of absolute values of its lower evaluated terms is less than

$$C_0 \frac{B^d - 1}{B - 1} < \tfrac{1}{2} B^d.$$

Consequently all three $D_j(B) > 0$, and $G_c > 0$ for $U_c \geq 0$. This proves the pre-binary positivity without inferring it merely from positive inputs.

The kernel degree bounds also give

$$|G_c| < 2C_0(1 + 3b + 9b^2)B^H \leq 26 C_0 b^2 B^H, \qquad 2G_c < N_1/64.$$

Since $Q = 1939 < H$, both A_T and $(B-2)A_T$ are less than $N_1/64$. It follows that

$$0 < J_c < N_1/32, \qquad J_c + (B-2)A_T + 2 < N_1.$$

The digits of M_0 are $B-1$, except that the 31 designated cells have digit $B-b$. Therefore $0 \leq M_0 < N_0$, and $g < N_0$ as well. The preceding strict high-block bound proves

$$0 \leq S_c, \quad 0 \leq T_c, \qquad S_c + T_c < N. \tag{2}$$

All source weights are positive, so $M_0 \geq B - 1$ and $M_0 - 2 \geq B - 3 > 0$. Moreover $S_c \geq N_0$, because J_c is a positive integer and $g \geq 0$. Hence

$$X = S_c(N-1) + T_c N + N - 1 > N_0(N-1) > 3b.$$

The reduced Y still exceeds $\max(b, 256)$. Because κ is even, $B \equiv N \equiv 1 \pmod 3$, $M_0 \equiv 0$, and $T_c \equiv -24$; hence $X \equiv T_c \equiv 0 \pmod 3$. These are all established before binary-base recovery.

6.3 Borrow-tolerant extraction and exact mask population

Suppose now that b is a power of two. So are B, N_0, N. The low mask uses $M_0 - 2$ instead of M_0. This clears precisely bit 1, since the zero-th base-B cell of M_0 is $B-1$. After recovery of a binary square base, $b \geq 4$ and $g = bg_0$ is divisible by four, so its bit 1 is zero already. Thus $g \,\&\, (M_0 - 2) = 0$ is equivalent to $g \,\&\, M_0 = 0$, which is exactly

$$g = \sum_i n_i B^{w_i}, \quad 0 \leq n_i < b.$$

For these digits put $p_i = n_i + 1$, so $U_c = \sum_i p_i B^{w_i}$ encodes positive parent coordinates without expanding a successor substitution in every residual. Write $h_s = [Z^s]G_p(Z)$ and let ϵ_s be the indicator of the 22 targets. All coefficients, including unsupported collisions, satisfy

$$|h_s| < C_0\{(\sum_i p_i)^2 + \sum_i p_i + 1\} \leq 993 C_0 b^2 < B/4.$$

The signed expansion of J_c is $\sum_{s=0}^{H}(2h_s + \epsilon_s)B^s$. Euclidean digit normalization has incoming carries only in $\{-1, 0\}$:

$$u_0 = 0, \quad u_{s+1} = \left\lfloor \frac{2h_s + \epsilon_s + u_s}{B} \right\rfloor, \quad d_s = 2h_s + \epsilon_s + u_s - Bu_{s+1}.$$

At a marked cell the mask $B - 2$ admits precisely digits $0, 1$. If $h_s = 0$, the unnormalized digit $1 + u_s$ passes. If $h_s \geq 1$, it lies in $[2, B)$ and fails. If $h_s \leq -1$, it lies strictly between $-B + 1$ and zero, and its normalized digit is at least two, so it fails. Thus incoming borrows do not affect the zero test. Positivity of J_c was proved independently, ensuring these are its ordinary digits. Combining the disjoint low and high blocks with (1) proves

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B^{w_i},\ 0 \leq n_i < b, \quad r_1(n+1) = \cdots = r_{22}(n+1) = 0.$$

For $N = 2^u$ and $S, T \geq 0$ with $S + T < N$, let $X_* = (S+T)N + N - 1 - S$. Its disjoint binary blocks give

$$s_2(X_*) = u + s_2(T) - \{s_2(S) + s_2(T) - s_2(S+T)\}.$$

The braces count carries with multiplicity, and vanish precisely when $S \,\&\, T = 0$. Legendre's formula therefore implies

$$2^{u+s_2(T)} \mid \binom{2X_*}{X_*} \quad \Longleftrightarrow \quad S \,\&\, T = 0.$$

The stronger range (2) is essential. Writing $b = 2^k$, $B = 2^v$, the actual mask population is

$$s_2(T_c) = (109 + 22)v - 31k - 23, \qquad u = 2064v.$$

Hence $u + s_2(T_c) = 2195v - 31k - 23 = \log_2 Y$, exactly explaining the fixed polynomial threshold used above.

Conversely, a positive parent zero is not the all-one vector: original row five equals one there. For arbitrarily large $b = 4^s$ exceeding all its coordinates, encode $n_i = p_i - 1$. Then $g \geq B > b$ and $g < \mathcal{C}$; the latter follows from $(b-1)\ell < bB^{108}$. All marked tests pass and $Y \mid \binom{2X}{X}$. This proves both directions of the entire source contract needed by the inherited finalizer, including its lower bound on forward witnesses and its pre-binary reverse bounds.

6.4 The Sun bridge registers

Set

$$L = lY, \quad U = 2LX, \quad V = 4gwY, \quad W_* = bw,$$
$$A = U(V+1), \quad B_* = 2X+1, \quad C = B_* + (A-2)h,$$
$$P = U^2V, \quad K = X + 1 + k(P-2), \quad S = 2A-5, \quad T = 3W_*C - 2(W_*^2-1),$$
$$Q = \big((P/2)^2 - 1\big)K^2 + 1,$$
$$\Delta = 8\mathcal{C}^3gK^2 - g^2\big(32(C-KL)^2\mathcal{C}^3 + g^2K^2\big).$$

The notation $P/2$ is only an abbreviation for the division-free expression $(LX)UV$, as $U = 2LX$; `Phalf` is exactly that expression in the circuit. Thus $4Q = (P^2-4)K^2 + 4$, the old auxiliary square radicand.

Put $\xi = zx$, and form Sun's modified Lucas certificate

$$D = (A^2-4)C^2 + 4, \qquad E = C^2D\xi, \qquad F = 4(A^2-4)E^2 + 1,$$
$$G_* = 2(A+2)(A-2)^2E^2 - 1, \qquad \Omega = 16E,$$
$$H_* = C + (B_*-C)F + \Omega Fy, \qquad I_* = (G_*^2-1)H_*^2 + 1.$$

Finally set

$$\rho = T + Sy, \quad \Sigma = \xi S^5, \quad \Theta = (\rho^3 - \rho S^2 - S^3)(\rho^2 + 23S^2).$$

The external bridge input is precisely Sun (2021), Theorems 4.12–4.13, specialized to $p = P_{\text{theorem}} = 2, Q_{\text{theorem}} = 1$, with the modified Lucas certificate and residue interpolation of Sun (2026), Theorems 3.1 and 4.1 and Propositions 6.1–6.2. These give the following interface.

- From a source witness with $b \in 2\uparrow$, $0 < g < \mathcal{C}$, and the central-binomial divisibility, choose positive l, w and integers h, k so that Q is a square, $S \mid T$, $\Delta > 0$, $C = u_{2X+1}(A)$, and $K = u_{X+1}(P)$.
- Conversely, if b is square, $f^2l^2\Delta > 0$, $\xi \neq 0$, $4Q$ and DFI_* are squares, and $S \mid T$, the bridge recovers $b \in 2\uparrow$, $0 < g < 2\mathcal{C}$, and the binomial divisibility.
- On the forward branch, ξ can be made a multiple of any prescribed positive integer, and y can be assigned any residue modulo ξ while retaining the square certificate DFI_*.

The exact forward values of C, K are not an extra assumption: the proof of Sun's 2021 Theorem 4.12 defines $C = u_B(A, 1)$ immediately before equation (4.16), which defines $K = u_{QX+1}(U^PV, 1)$. The recurrence used here is $u_0(a) = 0, u_1(a) = 1, u_{n+1}(a) = au_n(a) - u_{n-1}(a)$.

6.5 Congruences valid for every integer assignment

These identities do not assume positivity, a source solution, a Pell equation, or a zero of the final polynomial.

First, B, N_0, N are multiples of b^2. Therefore

$$X + 1 \equiv -g \equiv 0 \pmod{b}.$$

Also X is odd, because b, g, B, N_0, N are even. Since $b \equiv \beta_S \equiv 1 \pmod 3$, all powers of B equal 1 modulo 3. There are 22 targets. Modulo 3 one has $M_0 = 0$, $T_c = -24 = 0$, and hence

$$X \equiv (S_c + T_c) + 1 - 1 - S_c = T_c = 0 \pmod 3.$$

The displayed formula for Y has a factor b; hence $2b \mid U$ and $2b \mid A$. Consequently $b \mid P$ and $3 \mid P$. The normalization $k = b(3k_0 + 2)$ gives

$$K \equiv 0 \pmod{b}, \qquad K \equiv 1 + 2(-2) = 0 \pmod{3}.$$

It follows that

$$Q \equiv 1 \pmod{b^2}, \qquad Q \equiv 1 \pmod{3}, \qquad \gcd(b, Q) = 1.$$

In particular $Q \neq 0$, Q is odd, and for $r = bQ$ we have $r \equiv 1 \pmod{3}$, $|r| \geq 2$.

Since $h = bh_0$,

$$C + 1 = 2(X + 1) + (A - 2)bh_0 \equiv 0 \pmod{2b}.$$

Consequently $4b \mid D$: both terms of $D = A^2C^2 - 4(C^2 - 1)$ are divisible by $4b$.

6.6 The equation and unconditional two-root separation

Define

$$r = bQ, \quad a = r + (3\Delta - 1)(rz)^2, \quad d = DFI_*, \quad Z = \Sigma m - \Theta.$$

The single equation is

$$\boxed{F_{10} = \left(Z^2 - \Sigma^2(a + d)\right)^2 - 4\Sigma^4 ad = 0.}$$

The crucial separation is small and unconditional. Namely,

$$\frac{a}{b} = Q + (3\Delta - 1)bQ^2z^2 \quad \text{is odd}, \qquad \frac{d}{b} \quad \text{is even}.$$

Both displayed quotients are integers; they are not operations in the emitted polynomial. Since $b \neq 0$, this proves $a \neq d$ on every assignment. No magnitude comparison or large separating coefficient is required.

The following elementary lemma explains why this is enough. If integers $a \neq d$ and a rational number q satisfy

$$(q^2 - a - d)^2 = 4ad,$$

then $q \neq 0$, because at zero the difference is $(a - d)^2$. Define $u = (q^2 + a - d)/(2q)$ and $v = q - u$. Expanding gives $u^2 = a, v^2 = d$. A rational square root of an integer is integral, so both a, d are integer squares and $q = u + v \in \mathbb{Z}$. This proof includes negative and zero candidate radicands without making any prior choice of real or complex square roots.

Now $S = 2A - 5$ is odd and never zero. The polynomial $\pi(v) = (v^3 - v - 1)(v^2 + 23)$ has no rational root, so $\Theta = S^5\pi(\rho/S) \neq 0$. If $\Sigma = 0$, the final polynomial is $\Theta^4 \neq 0$. At any zero we therefore have $\Sigma \neq 0$, and in particular $\xi \neq 0$, $z \neq 0$. Apply the lemma with $q = Z/\Sigma$. It yields

$$a, d \in \mathrm{Sq}(\mathbb{Z}), \qquad \Sigma \mid \Theta.$$

The latter follows from $q = m - \Theta/\Sigma \in \mathbb{Z}$.

The standard quintic merger then gives $S \mid T$. For clarity: if for some prime p one had $v_p(T) < v_p(S)$, then $v_p(\rho) = v_p(T)$. The two factors of Θ have valuations $3v_p(T)$ and $2v_p(T)$, respectively. Thus $v_p(\Theta) = 5v_p(T) < 5v_p(S)$, contradicting $S^5 \mid \Theta$.

6.7 The combined square and positivity guard

Factor

$$a = r\{1 + (3\Delta - 1)rz^2\}.$$

The two factors are coprime. Neither is zero: $|r| \geq 2$, $z \neq 0$, and the second factor is 1 modulo r. Since their product is a square, each is a signed square and both have the same sign. But $r \equiv 1 \pmod 3$ excludes a negative square. Hence r and the bracket are positive squares.

Similarly $r = bQ$, $\gcd(b, Q) = 1$, and $b \equiv Q \equiv 1 \pmod 3$. Each factor is a signed square, and the same modulus excludes negative squares. Thus b, Q are positive squares. Since b is even and nonzero, $b \geq 4$, and therefore $r \geq 4$.

If $\Delta \leq 0$, then

$$1 + (3\Delta - 1)rz^2 \leq 1 - r \leq -3,$$

contradicting positivity of the bracket. Therefore $\Delta > 0$. Since $\mathcal{C} = bB^{108} > 0$, the formula for Δ gives $K \neq 0$ and $g > 0$. If $l = 0$, then $U = P = 0$ and $Q = 1 - K^2 \leq 0$, contradicting its positive-square value. Thus $l \neq 0$, and $f = 2t + 1 \neq 0$ was automatic. We have recovered $f^2 l^2 \Delta > 0$, exactly the old bridge hypothesis, without placing the factor $f^2 l^2$ in the new guard.

Before applying the bridge, rewrite

$$\Delta = gK^2(8\mathcal{C}^3 - g^3) - 32g^2(C - KL)^2\mathcal{C}^3 > 0.$$

Since $g > 0$ and $K \neq 0$, this gives $g^3 < 8\mathcal{C}^3$, hence $0 < g < 2\mathcal{C}$. The pre-binary source bounds already proved now yield $X > 3b$ and $Y > \max(b, 256)$ independently of any bridge conclusion. All hypotheses of the reverse bridge now hold. It yields the binomial source contract, which decodes a positive zero of the pinned parent. This proves soundness for all signed integer assignments, including degenerate values of the individual bridge variables.

6.8 Completeness of the new congruence restrictions

Suppose the parent has a positive zero. Choose a sufficiently large $b = 4^s$, $s \geq 1$, in the cofinal source construction. Then $t = (b - 4)/6 \in \mathbb{Z}$, and the encoded g is a multiple of b^2. In fact

$$X + 1 \equiv -g \equiv 0 \pmod{b^2}.$$

Apply the constructive bridge. It gives the exact values $C = u_{2X+1}(A)$, $K = u_{X+1}(P)$ and positive l, w, as stated above. We show its h, k already have the imposed forms.

The index $X + 1$ is even. The recurrence shows that $u_{2j}(P)$ is divisible by P. Since $2b \mid U$, we have $4b^2 \mid P$, and thus $b^2 \mid K$. Since $P \equiv 0 \pmod 4$, $v_2(P - 2) = 1$. In

$$(P - 2)k = K - X - 1,$$

the right side is divisible by b^2. As b is a power of two, $b^2/2 \mid k$, hence $b \mid k$. Also $P \equiv X \equiv 0 \pmod 3$ and $K \equiv 0 \pmod 3$, so $k \equiv 2 \pmod 3$. Because $b \equiv 1 \pmod 3$, the integer k/b is 2 modulo 3. It is exactly of the form $3k_0 + 2$.

For the other restriction, the recurrence gives $u_{2j+1}(A) \equiv (-1)^j \pmod{A^2}$. Since X is odd, $C + 1$ is divisible by A^2, hence by $4b^2$. Also $B_* + 1 = 2(X + 1)$ is divisible by $2b^2$. Thus $2b^2 \mid C - B_*$. The constructive A is a multiple of four, so $v_2(A - 2) = 1$. Therefore

$$h = (C - B_*)/(A - 2)$$

is divisible by b^2, in particular by b; write $h = bh_0$. All normalization restrictions thus preserve completeness, with no change to the constructive l, w, C, K.

These witnesses have $\Delta > 0$, and b, Q positive squares. Set $r = bQ$, also a positive square. The positive integer $\delta = 3\Delta - 1 \equiv 2 \pmod 3$ is nonsquare. Hence δr is a positive nonsquare. Pell's theorem gives integers $u, z > 0$ with

$$u^2 - \delta r z^2 = 1.$$

It follows that $a = ru^2$ is a square.

It remains to use the two free certificate variables without losing this chosen z. Since $S \mid T$, write $q = T/S$. First apply the residue interpolation theorem with prescribed divisor z to obtain its positive ξ. Then choose a residue class of y for which $\pi(q + y) \equiv 0 \pmod \xi$. The polynomial π has a root modulo every positive integer, so this residue exists. Interpolation supplies the square $d = DFI_*$, and $x = \xi/z$ is integral. Then $\Theta = S^5\pi(q + y)$ is divisible by $\Sigma = \xi S^5$. Choose

$$m = \Theta/\Sigma + \sqrt{a} + \sqrt{d} \in \mathbb{Z}.$$

The polynomial identity for F_{10} now gives $F_{10} = 0$.

For the local-root fact used here, the cubic $v^3 - v - 1$ has discriminant -23. At an odd prime other than 23, either it has a simple linear factor, or its irreducibility makes the discriminant a square modulo the prime, so $v^2 + 23$ has a simple root. At 23 the cubic has simple root 3. At 2, $-23 \equiv 1 \pmod 8$ is a square in $\mathbb{Z}_2$. Hensel lifting and the Chinese remainder theorem prove the assertion. The cubic has no rational root by the rational-root theorem, and neither does $v^2 + 23$. These are exactly the inherited quintic merger facts.

6.9 Circuit and exact degree certificate

The artifact `generated/sun10.json` is a division-free circuit with ten unrestricted integer unknowns and 519 live binary arithmetic gates: 330 multiplications, 163 additions and 26 subtractions. Fixed coefficient construction is excluded from this arithmetic metric. The decisive degree ledger is

R	X	Y	A	C	K	Q
$\deg R$	8254	4359	16976	16978	29592	118364
R	Δ	a	d	Σ	Z	F_{10}
$\deg R$	68559	305291	1494014	84882	84900	3327556

The leading term is $\Sigma^4 d^2$. Exact full-diagonal leading residues at degree 3327556, modulo $101, 1009, 1000003$, are respectively $19, 250, 375721$. Any one nonzero residue proves that this upper degree is attained.

The supplied checks establish all 22,544 source-isolation interactions, 58 local circuit identities, the factored Lucas identity, the two-root norm, actual coefficient bindings, gate liveness, and the degree certificate. Additional finite borrow, population, source-size, congruence, and Lucas-recurrence tests supplement the universal arguments; they do not replace the imported arithmetic or compiler theorems. Both ordinary Python and optimized Python use explicit failures rather than disabled assertions.

Against our previous degree-separated ten-integer circuit, the degree falls from 8469528 to 3327556 and the operation count from 533 to 519. Against the incoming congruence-norm circuit, the degree falls from 9152284 to 3327556 and the count from 586 to 519. These comparisons do not claim global optimality.

Provenance and dependency boundary. The new tail is adapted from `con_zfc_congruence_norms_2026-10-23`, received in commit `7331974b696693a8eed2aa5f0f64f1c19d9569f0`; its complete proof is retained under `upstream/`. The source encoding is the newly certified split108 placement with a global sign reversal of the two-shear source. The prior split116 source proof is preserved for provenance only; the formulas above and active layout supersede its weights and kernels. The complete original

four-port tuple is retained. Sun's 2021 Theorems 4.12–4.13 and 2026 Theorems 3.1, 4.1 and Propositions 6.1–6.2 remain explicit mathematical dependencies; the pinned source compiler theorem remains a dependency as well. This is not a new proof-assistant verification of the composite endpoint.

7 Thirteen integers without the 2026 Sun reduction

There is an explicit polynomial in thirteen unrestricted integer unknowns of exact degree 10 461 744, evaluated by 519 binary arithmetic operations, whose absence of integer zeros is equivalent in ZFC to Con(ZFC). This is a degree/cost tradeoff with the preceding 11 543 632-degree, 518-operation form, not a strict operation-count improvement. The unchanged finalizer uses the Jones ratio theorem, not Sun's 2026 reduction. The new source uses the split108 placement and a lower off-target positivity pad. A necessary new parity adjustment makes the square threshold cubic in ϵ rather than linear.

7.1 Degree-separated kernels and the complete polynomial

Use the original package-31 system of 22 quadratic residuals $r_j(p)$ in 31 strictly positive coordinates. The exact data are `split/parent_source.json`; the coherent fixed port tuple is $(\beta, K_0, M_C, M_F) = (\beta, K, M_C, M_{F,\mathrm{source}})$ from `FIXED_BINDING.json`. All four compressed little-endian payloads are included and hash-authenticated. The inherited compiler theorem identifies its positive common zeros with finite syntactic ZFC refutations. This is the original tuple, not a mixture with a later packet's smaller coefficients.

Perform just two elementary integer row operations:

$$\varrho_{14} = -(r_{14} + r_2), \qquad \varrho_{15} = -(r_{15} + r_3), \qquad \varrho_j = -r_j \quad (j \neq 14, 15).$$

First negate all rows and then undo the two shears, so the common zero set is unchanged. Write $\varrho_j = \sum_\alpha r_{j,\alpha} p^\alpha$ and

$$c_{j,\alpha} = \begin{cases} 2r_{j,\alpha}, & |\alpha| = 0, 1, \\ (\prod_i \alpha_i!) r_{j,\alpha}, & |\alpha| = 2. \end{cases}$$

Use the 31 positive weights w_i and the 22 targets q_j in `split/split_code.json`, and set

$$D_d(T) = \sum_{j,\,|\alpha|=d} c_{j,\alpha} T^{q_j - \alpha\cdot w} \quad (d = 0, 1, 2).$$

The supplied exact certificate establishes

$$W = \max_i w_i = 108, \quad Q_* = \max_j q_j = 1939, \quad (\deg D_0, \deg D_1, \deg D_2) = (1939, 1842, 1738).$$

The leading coefficients of (D_0, D_1, D_2) are $(2, 2, \beta)$; every kernel has internally distinct exponents. There are 89 source records. Checking all 22,544 same-degree source/expansion interactions proves

$$[T^{q_j}]\big(D_2(T)U(T)^2 + D_1(T)U(T) + D_0(T)\big) = 2\varrho_j(p), \qquad U(T) = \sum_i p_i T^{w_i}.$$

The degrees are kept in separate products. Thus a linear label may equal a quadratic label without causing cross-degree contamination. No homogeneous dummy coordinate or full Sidon condition is needed. Let

$$K_* = \max_d(\deg D_d + dW) = 1954, \qquad E_* = K_* + W + 2 = 2064.$$

The fixed coefficient bound, checked from the actual original port bit lengths, is $|c_{j,\alpha}| < A_* = 2^{2134927}$.

The unknowns are exactly

$$(\epsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, r, v) \in \mathbb{Z}^{13}.$$

All following expressions are polynomial abbreviations. Fix

$$\kappa = 2134940, \qquad a = 2^{\kappa} = 8192A_*, \qquad \eta = 2^{18146981} = 2^{17\kappa/2-9},$$

and define

$$\begin{aligned}
b &= 2\epsilon^2, & B &= ab^2,\\
\ell &= \sum_i B^{w_i}, & T_* &= \sum_j B^{q_j},\\
U_c &= \ell + g, & \mathcal{G} &= D_2(B)U_c^2 + D_1(B)U_c + D_0(B),\\
\mathcal{G}_+ &= \mathcal{G} + B^{1946}, & N_0 &= B^{109},\\
L &= B^{2064}, & \widehat{N} &= \eta\epsilon^3 B^{1089},\\
c &= bB^{108}, & G_c &= 32c,\\
M_c &= N_0 - 1 - (b-1)\ell, & J_c &= 2\mathcal{G}_+ + T_*,\\
S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)T_*,\\
R &= L^2 + (S_c + T_c)L + (L - 1 - S_c).
\end{aligned}$$

The extra monomial B^{1946} lies above every target. It repairs positivity on arbitrary signed g without changing an extracted residual or adding a new unknown. Every exponent in these formulas is fixed and nonnegative.

The circuit's register named N means the signed polynomial $\widehat{N}$. In the proof only, let $N = |\widehat{N}|$. Every occurrence of this parameter in the equation is through $N^2 = \widehat{N}^2$, so no absolute-value operation occurs in the polynomial.

Use the Jones registers with the exact CRT reparameterization of H below, and $\omega = w^2$:

$$\begin{aligned}
U &= N^2\omega, & Y &= N^2 s,\\
M &= RY, & P &= 2M^2U,\\
K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R + 1, & W_1 &= b\omega,\\
C &= B_1 + W_1 + \phi, & J &= 4A - 5, & Q &= 3W_1C - 2(W_1^2 - 1),\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= C + F(B_1 + jC), & I &= (G^2-1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,\\
u &= J, & t &= Q.
\end{aligned}$$

Set

$$\delta = C - KY, \qquad \Delta = g(c-g)K^2 - c^2\delta^2, \qquad (d_1, d_2, d_3) = (3s-1, 3\phi+2, 3\Delta-1).$$

The new coprime-spectator stack is

$$C_0 = a_0, \qquad B_j^* = 1 + 4d_jC_{j-1}z_j^2, \qquad C_j = C_{j-1}B_j^* \quad (j = 1, 2),$$

$$B_3^* = 1 + 4d_3C_2a_1z_3^2, \qquad C_* = C_3 = C_2B_3^*.$$

Thus the last guard contains the second square candidate a_1 as a spectator. This is the only change to the Pell sign stack. Finally define

$$\begin{aligned} Z &= \epsilon w i z_1 z_2 z_3 a_0, & \Sigma &= Z u^5, & \rho &= t + ur, \\ \Theta &= (\rho^3 - u^2\rho - u^3)(\rho^2 + 23u^2), & X &= \Sigma v - \Theta, & & \\ A_0 &= \Sigma^2 C_*, & A_1 &= \Sigma^2 a_1. & & \end{aligned}$$

The requested thirteen-integer equation is the following single polynomial:

$$\boxed{\mathcal{F}_{13} = (X^2 - A_0 - A_1)^2 - 4A_0A_1 = 0.} \tag{3}$$

The literal division-free DAG is `generated/signed_coprime_compact_split_crt.json`.

7.2 Signed-input bounds and exact square threshold

Assume only $\epsilon \neq 0$, hence $b \geq 2$ and $B = 8192A_*b^2$. Since all lower coefficients are bounded by A_* and the leaders are $(2, 2, \beta)$, geometric summation gives

$$D_0(B) > 0, \qquad 0 < D_1(B) < 3B^{1842}, \qquad D_2(B) > (\beta - 1)B^{1738}.$$

Here $\beta = 2^{1064921} - 1 > 10$. Completing half the quadratic square, for every real U_c,

$$\mathcal{G}_+ \geq \tfrac{1}{2}D_2(B)U_c^2 + B^{1946} - \frac{D_1(B)^2}{2D_2(B)} > \tfrac{1}{2}D_2(B)U_c^2 + \tfrac{1}{2}B^{1946},$$

because the subtracted term is less than $9B^{1946}/(2(\beta - 1)) < B^{1946}/2$. Thus for every integer g, writing $U_c = \ell + g$,

$$S_c > N_0D_2(B)U_c^2 + U_c + N_0B^{1946} - \ell > 0.$$

The first two terms have nonnegative sum for integer U_c, and $\ell < N_0$. The mask gives $T_c > 0$ as well. Therefore $R > L^2$ before any sign guard or source decoding.

The new threshold is not the old linear-ϵ expression. It satisfies the exact identity

$$\widehat{N}^2 = 2^{-21}a^{17}b^3B^{2178} < B^{2195} < B^{4128} = L^2.$$

Also $N = |\widehat{N}| = \eta|\epsilon|^3B^{1089} \geq bB^{108} = c$, since $\eta|\epsilon|B^{981} \geq 2$, and $N \geq 8$. Thus the inherited ratio hypotheses $R \geq N \geq 8$ and $b, c \leq N$ hold before signs of the other free variables are recovered.

For $0 < g < c$, $U_c < 2bB^{108}$ and the old coefficient-majorant argument gives

$$|\mathcal{G}| < 12A_*b^2B^{1954}, \qquad 2|\mathcal{G}| < B^{1955}/256, \qquad T_* < B^{1955}/256.$$

The extra pad B^{1946} is smaller than B^{1954}, so $0 < J_c < B^{1955}/64$. Consequently $S_c < L/32$ and $T_c < L/64$, in particular $S_c + T_c < L$. The pad is above every target, since $1946 > 1939$, and below the convolution height. It changes no tested residual. All unnormalized coefficients of $\mathcal{G}_+$ remain below $B/8$ in absolute value under the inherited $994A_*b^2$ bound.

Exactly the sparse-borrow argument gives

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_iB^{w_i}, \quad 0 \leq n_i < b, \quad r_j(n + \mathbf{1}) = 0 \ (1 \leq j \leq 22).$$

Indeed the unnormalized target digit is $4\varrho_j(n + \mathbf{1}) + 1$ and the incoming borrow is 0 or -1; it avoids the mask $B - 2$ exactly when that residual vanishes. Non-target collisions cause no problem. The source all-one vector is not a zero, so the forward encoded g is positive. Cofinal choices $b = 2^{1+2r}$ accommodate every positive source tuple.

After the Jones theorem recovers $b = 2^k$, the identity $b = 2\epsilon^2$ forces odd k. Put $B = 2^\nu$, $\nu = \kappa + 2k$. The actual mask population is $131\nu - 31k - 22$, and the tight source range yields

$$s_2(R) = 2195\nu - 31k - 21 - \tau_2(S_c, T_c).$$

The cubic-ϵ threshold is calibrated exactly:

$$2195\nu - 31k - 21 = 2(17\kappa/2 - 9 + 1089\nu) + 3(k-1) = \log_2 \widehat{N}^2.$$

The exponent $3(k-1)$ comes from ϵ^6, not ϵ^2; retaining the previous linear-ϵ threshold would be incorrect for this source parity. Legendre's identity therefore proves $N^2 \mid \binom{2R}{R}$ exactly when the masks are disjoint. The following retained finalizer proof now applies without any other source change.

7.3 An exact polynomial CRT reparameterization

The inherited ratio interface uses $H_{\text{old}} = B_1 + j_{\text{old}}C$ and $JF \mid QF + J(H_{\text{old}} - C)$. Since $E = iJDC^2$,

$$F \equiv 1 \pmod{J}, \qquad F \equiv 1 \pmod{C},$$

and hence $\gcd(J, F) = 1$. The divisibility is equivalent to $J \mid Q$ and $F \mid H_{\text{old}} - C$. Our changed register

$$H = C + F(B_1 + jC)$$

automatically satisfies the latter. It also has the original form, with

$$j_{\text{old}} = 1 + (A^2 - 1)i^2J^2D^2C^3B_1 + Fj,$$

which is a polynomial integer, since $F - 1 = (A^2 - 1)i^2J^2D^2C^4$. This identity holds for arbitrary signed inputs, including $C = 0$.

Conversely, at the canonical positive ratio witnesses, $0 < B_1 < C$ and $j_{\text{old}} \geq 1$. Therefore $z = (H_{\text{old}} - C)/F > 0$ and $z \equiv B_1 \pmod{C}$, so $z = B_1 + jC$ with $j \geq 0$. Thus every canonical witness is represented. No extra unknown or cofinal restriction is introduced. The merger now has $u = J$ rather than JF. This CRT step is inherited from the separately identified sparse-Pell packet, not claimed as an original theorem here.

7.4 The interval guard and the Jones interface

If $\Delta > 0$ and $b \geq 2$, then $K \neq 0$ and $0 < g < c$. The identity $4g(c-g) = c^2 - (c-2g)^2 \leq c^2$ implies $4\delta^2 < K^2$. Along with $s \geq 1$, $\phi \geq 0$, and $w \neq 0$, this gives $C, Y > 0$ and $K > 0$. The usual square-gap argument forces $h \geq 1$: if $h \leq 0$, then $1 \leq K \leq R + 1 < P$; $K = 1$ would force $P - 1 \mid R$, which is impossible, while for $2 \leq K < P$,

$$(PK - 1)^2 < (P^2 - 1)K^2 + 1 < (PK)^2.$$

Thus a_0 cannot be a square in that branch.

For precision, instantiate the free ceiling in `Jones1982.Nine.ratio_sound` by $G_{\text{aux}} = 8gK^2 + 1$, rather than assuming its old margin condition. The positive integer $K^2 - 4\delta^2$ then gives

$$(K^2 - 4\delta^2)G_{\text{aux}} - 8gK^2 > 0.$$

This ceiling is a proof-side abbreviation; the ratio registers do not otherwise depend on it.

Replace i by $|i|$, which changes no condition. To handle signed j, $u \mid t$ now means $J \mid Q$, while $F \mid H - C$ holds identically. Also $F \equiv 1 \pmod{D}$, $G \equiv 1 \pmod{D}$, and $G \equiv A \pmod{F}$ show that the positive integers D, F, I are pairwise coprime. Hence $a_1 = DFI$ square makes each a square. Put

$d = G^2 - 1 > 0$ and $x = \sqrt{I}$. The norm-one unit $x + H\sqrt{d}$ is positive even if $H < 0$. A positive power of $G + \sqrt{d}$ is the identity modulo CF; multiplying by sufficiently large such powers gives a Pell solution $H' > B_1$ with $H' \equiv H \pmod{CF}$. Then $j' = (H' - B_1)/C > 0$, all square conditions remain true, and $J \mid Q$ remains true. The literal positive Jones interface therefore applies and gives the power-of-two base and central-binomial condition required above.

For completeness, take any common source zero, choose $b = 2^{1+2r}$ large enough, and set $\epsilon = 2^r$. The direct source code has $32g < G_c$, so the literal ratio completeness theorem supplies its canonical witnesses. In `Ratio.lean`, `exists_B25core` sets

$$\omega = 2^{2R+1-k}, \qquad U = (N^2/b)2^{2R+1}.$$

Since $k = 1 + 2r$ is odd and $k < 2R + 1$, this ω is a positive square. No claim that the canonical s is a square is needed.

The same canonical construction has $\rho_0 = (U+1)^{2R}/U^R = Y + f$ with $0 \le f < 2^{2R}/U = b/(2N^2)$ and $Y \ge 1$. The source's `ratio_core` gives

$$|C/K - \rho_0| \le \frac{2(Y+f)}{Y(U+1)} < \frac{4}{U}, \qquad |C/K - Y| < \frac{b}{N^2}.$$

The last inequality uses $R \ge 8$. Since $c, b \le N$ and $g(c-g) \ge 1$, it follows that

$$\frac{c^2(C - KY)^2}{K^2} < \frac{c^2 b^2}{N^4} \le 1 \le g(c-g),$$

so the very same canonical witnesses, translated by the exact CRT formula, satisfy $\Delta > 0$. This is a cofinal square-base proof, not an unsupported restriction of an existing slack variable to squares.

7.5 Raw two-root recovery, including the zero spectator branch

For every integer assignment, $u = J = 4A - 5 \neq 0$. The polynomial

$$\Pi(x) = (x^3 - x - 1)(x^2 + 23)$$

has no rational root and has a root modulo every positive integer. For an odd prime other than 23, either the cubic has a simple root or it is irreducible, whose Frobenius three-cycle makes its nonzero discriminant -23 a square. The quadratic then has a simple root. At 23 the cubic has the simple root 3; at 2, $-23 \equiv 1 \pmod 8$ admits compatible square roots. Hensel lifting and CRT finish the assertion. Thus $\Theta = u^5 \Pi(t/u + r) \neq 0$ for every assignment, and

$$\exists r \quad Z u^5 \mid \Theta \quad \Longleftrightarrow \quad Z \neq 0 \text{ and } u \mid t.$$

Indeed a root modulo $|Z|$ proves the reverse implication once $u \mid t$. If $v_p(u) = \alpha > \beta = v_p(t)$, the two factors of Θ have valuations 3β and 2β, contrary to $u^5 \mid \Theta$. If $Z = 0$, divisibility would require $\Theta = 0$, already impossible.

At a zero of the equation, $\Sigma = 0$ is impossible because then the polynomial is $\Theta^4 \neq 0$. In particular $\epsilon, w, i, z_1, z_2, z_3, a_0$ are nonzero. Every $d_j \equiv 2 \pmod 3$ is nonzero, and every $B_j^* \equiv 1 \pmod 4$ is nonzero. Thus C_2, C_* are nonzero.

No coincident radicands. One has $C_* \neq a_1$ without any positivity assumption. If $a_1 = 0$ this follows from $C_* \neq 0$. Otherwise $B_3^* \neq 1$, and it cannot be -1 because it is 1 modulo 4. But

$$\gcd(B_3^*, a_1) = 1,$$

while $C_* = a_1$ would make B_3^* divide a_1. This would force $B_3^* = \pm 1$, a contradiction.

Put $q = X/\Sigma$. The raw norm equation implies

$$(q^2 + C_* - a_1)^2 = 4C_* q^2.$$

If $q = 0$, the norm would give $C_* = a_1$, just excluded. Therefore

$$r_0 = \frac{q^2 + C_* - a_1}{2q}, \qquad r_1 = q - r_0$$

are rational, with $r_0^2 = C_*$ and $r_1^2 = a_1$. Rational square roots of integers are integers. Hence C_*, a_1 are integer squares, $q \in \mathbb{Z}$, and $\Sigma \mid \Theta$. In particular $J \mid Q$. No complex-root separation assumption is used.

Recover the first two guards before ruling out $a_1 = 0$. The factors a_0, B_1^*, B_2^*, B_3^* are pairwise coprime, because each new factor is 1 modulo its entire preceding product. Their nonzero product C_* is a square. Thus each absolute factor is a square. Since $B_j^* \equiv 1 \pmod 4$, every B_j^* is positive: a negative square cannot be 1 modulo 4. The product sign then forces $a_0 > 0$, also a square. All prefixes are now positive, so $d_1, d_2 > 0$, yielding $s \geq 1$ and $\phi \geq 0$.

At this point the last factor might still equal 1 if $a_1 = 0$; this branch must not be silently discarded. The unconditional source estimate above has already proved $R > L^2 > 0$ for *every* signed g. Since $\epsilon, w \neq 0$ and $s \geq 1$, we have $b \geq 2$, $N \geq 8$, $U, Y > 0$, $A \geq 2$, and, using $\phi \geq 0$, $C > 0$. Then

$$D \geq A^2 > 0, \qquad F \geq 1, \qquad CD + 1 - A > 0, \qquad G \geq A > 1, \qquad I \geq 1.$$

Consequently $a_1 = DFI > 0$. Now $B_3^* \neq 1$, and its positivity with $C_2 a_1 z_3^2 > 0$ forces $d_3 > 0$, hence $\Delta > 0$. This closes the zero-spectator branch without adding a_1 to the nonzero multiplier. The interval and ratio arguments above therefore apply.

Completeness. Start from the canonical positive ratio witnesses, translated by the CRT parameterization. They have a_0, a_1 positive squares, $J \mid Q$, and all three strict sign conditions. For $j = 1, 2$, $d_j C_{j-1}$ is positive nonsquare because $d_j \equiv 2 \pmod 3$ and C_{j-1} is a square. A nontrivial Pell solution $x^2 - d_j C_{j-1} y^2 = 1$ can be chosen with y even, so $z_j = y/2 \neq 0$ makes $B_j^* = x^2$. For the last guard, $d_3 C_2 a_1$ is likewise positive nonsquare, since $C_2 a_1$ is a positive square. The same construction gives B_3^* square and C_* square. The intersective lemma chooses r with $\Sigma \mid \Theta$. Finally choose

$$v = \Theta/\Sigma + \sqrt{C_*} + \sqrt{a_1} \in \mathbb{Z}.$$

Then the raw norm vanishes. Both directions are proved. By the inherited compiler theorem, integer-zero existence is equivalent to a finite ZFC refutation, and ZFC proves its negation equivalent to Con(ZFC).

7.6 Exact degree and validation

The emitted DAG is `generated/signed_coprime_compact_split_crt.json`. It uses thirteen unrestricted variables and 519 live gates: 327 multiplications, 169 additions, and 23 subtractions. Fixed leaf construction is excluded from this arithmetic metric. The supplied four coefficient integers are unchanged.

The independent verifier propagates an upper degree and the leading coefficient on the full diagonal using the actual fixed coefficients. The leading residues at degree 10461744 are 20, 442, and 336577 modulo 101, 1009, and 1000003, respectively. Any one nonzero residue certifies that this upper degree is attained. It also replays all named registers against independent formulas on signed assignments, checks the source certificate, and tests the guard and CRT identities. The replay is finite supplementary evidence, not a re-proof of the imported unbounded compiler or Jones theorem. The full retained prior proof and provenance are included under `upstream/`; the active definitions and cubic threshold above supersede the prior source registers.

8 One free scale and an index absorber: fourteen positive variables

A useful new tradeoff is a polynomial of exact degree 30 in fourteen strictly positive integer unknowns, with 87 binary arithmetic operations. The comparison is the existing fourteen-positive degree-52, 79-operation construction: degree decreases, cost increases. It also saves one variable relative to the incoming integral-root degree-24/90, 26/88 and 30/79 constructions. No global minimality or domination is asserted.

The idea is to free *only* the ordinate scale Y using a monic-root carrier, while retaining $X = wq$. The recovered divisibility $q^3 \mid Y$ then permits the adaptive $U = 4qY$ index absorber to delete its quotient witness. Attempting to merge both scale offsets into a single square root does not imply their separate divisibilities, and is not used here.

8.1 Literal polynomial

Use the entire unchanged compact tuple from the Integral Root packet: $\beta = 2^{590069} - 1$, $K_0 = K$, M_C, and $M_F = M_{F,\mathrm{source}}$, with the latter already shifted. Its four actual integer payloads and binding are included. The fourteen positive unknowns are

$$J, F, \alpha, z, f, i, T, w, Y, r, \eta, \zeta, u, \gamma.$$

All remaining symbols are polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha,\\
X &= wq, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3,\\
\Delta &= a^2 + H, & \ell &= EkY,\\
A &= \ell(\ell + k) + 1, & G &= \Delta c^2 + 1,\\
U &= 4qY, & v &= k - R - 1,\\
Q &= \Delta(f^2 - 1), & I &= ic^2.
\end{aligned}$$

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
D_* &= U(X + ac) + H(U\gamma - v),\\
V &= f^2(cT - R) - c,\\
N_t &= (K_0 + w)C + q - F - z(q-1).
\end{aligned}$$

Define

$$\begin{aligned}
A_0 &= [q^3(\ell + r) + Y]^2 - q^6 A,\\
A_1 &= D_*^2 - U^2 G,\\
A_s &= I^2 - Q,\\
A_a &= (V + u)^2 - Qu(2V + u) - 1,\\
A_t &= N_t - 1.
\end{aligned}$$

The promised equation is

$$\boxed{P_{14} = A_0^2 + A_1^2 + A_s^2 + A_a^2 + A_t^2 = 0.}$$

There are no divisions, variable exponents or hidden existential variables in this literal expression. The five residuals can alternatively be used as a system of degrees $12, 14, 10, 15, 2$, with a shared 78-operation circuit.

8.2 Correctness of the new composition

At a zero, the rational square root $\tau = \ell + r + Y/q^3$ of the integer A is integral. Hence $q^3 \mid Y$; write $Y = sq^3$ with $s > 0$. This recovery precedes every application of an old scale-dependent bound. The transport equation gives the inherited whole-tuple bounds

$$q \geq 16, \quad C > 0, \quad 3q < R < q^4 - 2, \quad E \geq q^4 > R + 2.$$

The first Pell equation supplies $\tau = \chi_{P_0}(n)$ and $k = 2\psi_{P_0}(n)$, where $P_0 = 2XY^2 + 1$ and $n > 0$. The second residual makes $g = D_*/U$ integral. It is positive, because

$$D_* > U(X + ac) - Hk > ak(UY - 5) > 0.$$

Thus $g = \chi_{a+2}(p)$ and $c = \psi_{a+2}(p)$ for $p > 0$. The absorber gives $U \mid H(k - R - 1)$ without cancelling H. The adaptive modulus satisfies $U \geq 4q^4$, $U \mid 4XY^2$ and $\gcd(U, H) \leq 3$. For the last claim, $\gcd(Y, H) \mid 3$; every prime of q divides Y; and if $3 \mid q$ then $27 \mid Y$, so $H \equiv 3$ modulo 27. These are precisely the properties needed by the retained absorber proof. Moreover, $k \equiv 2n \pmod{4XY^2}$, with $U \mid 4XY^2$. Consequently

$$\mu = \frac{U}{\gcd(U, H)} > R + 1, \qquad 2n \equiv R + 1 \pmod{\mu}, \qquad n \geq (R + 1)/2 \geq 25.$$

The standard first/main Pell comparison gives $p > n$, and therefore $c > (a + 2)\Delta^2$, $c > 2p$ and $c > 2R$. These bounds justify the inherited *ordinary* strong-rank lemma for $(ic^2)^2 = \Delta(f^2 - 1)$, yielding a main Pell index m with $p \mid m$, $c \mid m$, $c^2 \mid f^2 - 1$ and $m \geq c > 2p$.

Set $y = V + u$. The auxiliary identity is exactly

$$QV^2 - (Q - 1)y^2 = 1.$$

Although V need not be positive, y is: the identity gives $|y| \geq |V|$, while $u = y - V > 0$ rules out $y < 0$; zero is impossible. With $T_{\text{old}} = fT$, the signed expression $V = c(T_{\text{old}} f - 1) - Rf^2$ is the inherited auxiliary expression. Its index proof gives $p = R$; the absorber then gives $n < R$ and $2n = R + 1$. Hence

$$h = \frac{k - R - 1}{E} > 0, \qquad \gamma_{\text{old}} = \gamma - \frac{k - R - 1}{U}$$

are integral and recover the original equations. The positivity of the latter is not inferred from integrality: $k \geq 4P_0 > X$, $c > k > X$ and $g > (a + 1)c$ imply $g - (X + ac) > c - X > 0$, so $\gamma_{\text{old}} > 0$. The inherited ordinary compiler therefore produces a finite ZFC refutation.

Conversely, a finite refutation supplies canonical compact-tuple witnesses. Convert the normalized strong ordinate by $i = \Delta i_0$, and use the absorber shift $\gamma = \gamma_{\text{old}} + (k - R - 1)/U > 0$. The Integral Root positive-gap lemma gives $\tau > \ell + s$, so $r = \tau - \ell - s > 0$. The retained squared-modulus completeness proof chooses T_{old} divisible by f and $y > V$, giving positive $T = T_{\text{old}}/f$ and $u = y - V$. This is an existential choice of canonical witnesses, not a same-coordinate identity for arbitrary old witnesses. All five residuals vanish. The imported compiler calibration and this composition prove in ZFC that P_{14} has a positive zero exactly when there is a syntactic ZFC refutation; negation yields the desired equivalence to Con(ZFC).

8.3 Degree and replay boundaries

Writing $Z = UX + H(U\gamma - v)$ gives exact identities

$$A_0 = q^6[r^2 + \ell(2r - k) - 1] + 2q^3(\ell + r)Y + Y^2,$$
$$A_1 = 2UacZ + Z^2 - U^2(Hc^2 + 1).$$

Thus their degrees are 12 and 14, not the naive uncancelled bounds. For the second, Z has leading term H_0R_0 of degree 7, where $R_0 = (\beta J)^3(\beta J - F)$; its term $H_0U_0\gamma$ has only degree 6. The auxiliary residual has degree 15, with leading form $2E_0^2R_0f^4u$, where $E_0 = w\beta JY$. Hence the unique degree-30 homogeneous part of P_{14} is

$$4w^4(\beta J)^{10}Y^4(\beta J - F)^2f^8u^2,$$

which is nonzero for the actual positive β. The primary circuit uses 47 multiplications, 24 additions and 16 subtractions; the cancellation certificate is a separate circuit, not the advertised evaluator.

The attached standard-library replay authenticates forty source pins and all four fixed integer payloads, verifies three universal sparse polynomial identities, checks liveness and costs, and compares independently coded literal formulas. Exact-degree specializations use the actual coefficients modulo three primes. Three isolated corruption controls are rejected, including under Python optimization. These are arithmetic and integrity audits, not a fresh proof-assistant endpoint; the infinite Pell and compiler inputs remain explicitly inherited mathematical proofs.

9 Beyond K215 and Jones: an architectural fork

The search need not preserve the blank-tape machine. A concrete next experiment is to use a standard fixed-input universal machine and place the ZFC-refuter program in a coefficient-supported initial boundary. Neary–Woods provide standard universal machines with pairs $(5,5),(6,4),(9,3),(15,2)$; the five-symbol machine uses 22 instructions (https://doi.org/10.3233/FI-2009-0036). This suggests reducing the native transition alphabet, not claiming a smaller polynomial from machine size alone.

The missing theorem is an economical finite-input boundary compiler. The input must not be expanded into a new tile per symbol, nor represented as a polynomial in a variable base whose degree is its length. A useful prototype first proves arbitrary finite input boundaries for a test machine, then materializes a universal machine's exact refuter input and counts all coefficients. No such endpoint is claimed here.

A genuinely independent published route is Carl–Moroz's direct Diophantine representation of first-order provability, specialized to finite Gödel–Bernays axioms (https://webdoc.sub.gwdg.de/ebook/serien/e/mpi_mathematik/2007/132.pdf). It removes proof enumeration but, in its published explicitization, is much larger than the present frontier. For a proof-oriented alternative, Larchey-Wendling–Forster's constructive Coq DPRM route through FRACTRAN is available (https://arxiv.org/abs/2003.04604); its extracted size for this refutation task remains to be measured.

A tempting shortcut is invalid: deleting FRACTRAN priorities and allowing any applicable fraction leaves a vector-addition system under prime valuations. Its ordinary reachability is decidable, so it cannot serve as a uniform universal refutation engine. An actual zero-test or equivalent capability must remain. Finally, every consistency calibration must be internal to ZFC and use finite proof translations, not an assumption of ZFC's arithmetic soundness. The accompanying `alternatives/REVIEW.md` records the exact open obligations.

10 Formalization boundary and validation

The recent `dabf4e39f103` commit reports local Lean verification of package52's literal60-gate/22-row source,136 shifted coefficient records, position/kernel bounds, sparse low-mask decoding, cofinal estimates, ratio substitution and a margin-first finalizer. The source adapter, actual native calibration, assembled closed replacement expression and internal ZFC equivalence are still expressly separate obligations. We read the reported3709-job build and axiom audit; we did not rerun that Lean build. Existing closed `ExplicitZFC` and `NativeZFC` endpoints are not automatically certificates for the smaller circuits in this archive.

The arithmetic proofs here establish new equivalences using precisely identified inherited compiler/Pell contracts. Their ordinary ZFC-formalizable status is distinguished from a checked external Lean theorem and from an exported object-language ZFC derivation. Only the ten-integer route additionally imports Sun's2026 result. No use is made of an assumption that ZFC is arithmetically sound or that either side of the consistency equivalence is true.

The complete release is hash-authenticated, copied to an isolated directory, and replayed without network access. Exact polynomial identities, source isolation, coefficient payloads, gate liveness, cancelled degree upper bounds and nonzero leading witnesses are checked. The two quartic projections have unit leading coefficients on the actual fixed slice. Component mutation tests are distinguished from root byte-integrity corruption controls. Normal and optimized Python runs are recorded separately; no disabled assertion run is silently reported as a proof.

The root receipt lists every executed stage and every regenerated file compared byte-for-byte. PDF generation is independent of the mathematical replay; PDF metadata can differ when recompiled. The distributed ZIP is extracted and the complete replay repeated before delivery. These finite checks support explicit arithmetic/data claims and implementation fidelity; they do not replace the universal proofs or certify an unbounded compiler by testing finitely many assignments.

Corrections and rejected shortcuts. The incoming Q45 prose writes a transport quotient shifted by one, while its literal Q45 and Q50 row is $(K_0 + w)C = F + z(q - 1)$. We retain the literal row; the corrected late argument gives $z > 0$, exactly what is needed. The original files are preserved and the correction is documented separately. The unsuccessful shared-sign proposal with only $w = u^2$ is not claimed; its negative branch lacked the early scale gap. Likewise the initial positive14 draft degree34 is superseded by the adaptive-modulus degree30 form, not counted as an additional frontier improvement. No unproved cofinal restriction on a canonical transport witness is used.

Attribution and reproducibility

The retained notices credit the native compiler, Pell and signed step-down lemmas, Jones and Sun interfaces, and each of the six incoming constructions. The direct-proof and universal-machine alternatives are literature-backed research directions, not numerical endpoints in the result table. Internal independent reviews are AI-assisted and do not represent external human peer review. All mathematical claims, exact artifacts and their limitations are kept together for incorporation into ProveIt.